

**The Faculty of Medicine of Harvard University
Curriculum Vitae**

Date Prepared: June 26th, 2025
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Place of Birth: China

Education:

09/2006	B.E.	Automation	Shanghai Jiaotong University, China
06/2016	PhD	Computer Science, Distinguished Prof. Tianming Liu	University of Georgia

Postdoctoral Training:

08/2016- 07/2019	Research Fellow	Medical Image Analysis, Assoc. Prof. Quanzheng Li and Distinguished Prof. James H. Thrall	Harvard Medical School and Massachusetts General Hospital
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Faculty Academic Appointments:

08/2019- 05/2023	Instructor	Radiology	Harvard Medical School
06/2023-	Assistant Professor	Radiology	Harvard Medical School

Appointments at Hospitals/Affiliated Institutions:

01/2007- 06/2009	Research Assistant	The College of Environmental Science and Engineering	Nankai University, China
08/2019-	Research Staff	Radiology	Massachusetts General Hospital

Major Administrative Leadership Positions:

International

2019, 2022- 2024	Founding program chair for the <i>International Workshop on Multiscale Multimodal Medical Imaging</i>	The Medical Image Computing and Computer-Assisted Intervention Society (MICCAI)
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2021	Organizer and program chair for the <i>International Workshop on Multimodal Learning and Fusion Across Scales for Clinical Decision Support</i>	The Medical Image Computing and Computer-Assisted Intervention Society (MICCAI)
2024	Organizer and program chair for the <i>SAGES Critical View of Safety Challenge</i>	The Medical Image Computing and Computer-Assisted Intervention Society (MICCAI)
2024	Organizer and program chair for the <i>NeurIPS Workshop on Advancements in Medical Foundation Models</i>	NeurIPS Foundation

Committee Service:

International

2015	Program Committee	MICCAI Workshop on Machine Learning in Medical Imaging
2017,2023	Program Committee	International Conference on Brain Informatics
2023	Program Committee	International Workshop on Medical Image Learning with Noisy and Limited Data
2023-2024	Area Chair	International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)
2025	Program Committee	IEEE Symposium on Computer-Based Medical Systems

Professional Societies:

2011-	Institute of Electrical and Electronics Engineers (IEEE), International	Member
2011-	Engineering in Medicine and Biology Society (EMBS), International	Member
2011-	The Medical Image Computing and Computer-Assisted Intervention Society, International (MICCAI), International	Member
2017-	American Roentgen Ray Society (ARRS), Regional	Member
2023-	American Medical Informatics Association (AMIA), Regional	Member
2023-	Society of Nuclear Medicine and Molecular Imaging (SNMMI), National	Member

Grant Review Activities:

2024	ZRG1 CTH-E (57): PAR Panel: Academic-Industrial Partnerships for Translation of Technologies 10/16/2024	National Institutes of Health, Division of Translational and Clinical Sciences (DTCS) Ad hoc reviewer
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2025	ZMH1 ERB-E (03) S: BRAIN Initiative: Research on the Ethical Implications of Advancements in Neurotechnology and Brain Science 05/02/2024	National Institutes of Health, BRAIN Initiative Ad hoc reviewer
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Editorial Activities:

• Ad hoc Reviewer

Human Brain Mapping
IEEE Transactions on Affective Computing
IEEE Transactions on Biomedical Engineering
IEEE Transactions on Emerging Topics in Computational Intelligence
IEEE Transactions on Knowledge and Data Engineering
IEEE Transactions on Medical Imaging
IEEE Transactions on Pattern Analysis and Machine Intelligence
JAMA Network Open
Medical Image Analysis
Nature Communications
Nature Human Behavior
Nature Medicine
Neuroimage
Neuroscience
Progress in Neurobiology

• Other Editorial Roles

2021-	Associate Editor	Frontiers in Oncology
2021-	Associate Editor	Frontiers in Radiology
2021-	Associate Editor	Frontiers in Neuroscience
2021	Guest Editor	Frontiers in Neuroscience, Special Issue “Multi-Dimensional Characterization of Neuropsychiatric Disorders”
2022-	Associate Editor	Frontiers in Cardiovascular Medicine
2023-	Associate Editor	Meta-Radiology
2023-	Associate Editor	BMC Biomedical Engineering
2024	Associate Editor	Connected Health And Telemedicine
2025	Guest Associate Editor	IEEE Transactions on Medical Imaging
2025	Associate Editor	Scientific Report

Honors and Prizes:

2011	Best Student Paper Award	IEEE International Symposium on Biomedical Imaging	“Brain State Change Detection via Fiber-centered Functional Connectivity Analysis”
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2013	Best Student Paper Award	IEEE International Symposium on Biomedical Imaging	“Discovering Common Functional Connectomics Signatures”
2015	Paul D. Coverdell Neuroimaging Franklin Foundation Scholars Program Travel Award	Department of Psychology, University of Georgia	
2015	Cover and Feature Paper	IEEE Transactions on Biomedical Engineering	“Holistic atlases of functional networks and interactions reveal reciprocal organizational architecture of cortical function”
2016	Outstanding Graduate Dissertation/Thesis	University of Georgia	
2020	Best Paper Awards	IEEE International Symposium on Biomedical Imaging	“ASCNet: Adaptive-Scale Convolutional Neural Networks for Multi-Scale Feature Learning”
2021	MGH Thrall Innovation Grants Award	Massachusetts General Hospital	“Chest Radiographs-based Lung Cancer Screening by the DeepProjection Technique”
2022	Best Paper Awards	IEEE Transactions on Radiation and Plasma Medical Sciences	“Deep Learning-Based Image Segmentation on Multimodal Medical Imaging”
2024	Research Scholar Program	Google Inc.	“Tailoring Large Language Models for the Diagnosis and Management of Late-life Depression Patients with Limited Access to Healthcare Resources”
2024	Distinguished Paper Award	American Medical Informatics Association (AMIA)	“MKRAG: Medical Knowledge Retrieval Augmented Generation for Medical Question Answering”

Report of Funded and Unfunded Projects

Past

9/21/2020-03/01/2021	“Pseudo Chest CT from Chest X-RAY, COVID-19 Workstream” GE Precision Healthcare, Industrial Grant Project Leader (PI: Quanzheng Li) Development of a machine learning model, “DeepProjection,” that can generate pseudo-CT from X-rays images to improve the diagnosis and management of patients with COVID-19. The model will apply to COVID-19 patients and synthesize chest computed tomography (CT) images from a chest X-ray in COVID-19.
06/01/2021-11/01/2022	“Aortic Stenosis Clinical Applications” GE Precision Healthcare, Industrial Grant Project Leader (PI: Quanzheng Li) Developing an application for the intelligent management of aortic stenosis (AS) that aims

to predict relevant clinical outcomes from AS patients undergoing surgeries based on electronic health records (EHR), radiological reports, and imaging data.

“Identification of Multi-modal Imaging Biomarkers for Early Prediction of MCI-AD Conversion via Multigraph Representation”

NIH R03, 1R03AG078625-01, Direct cost: \$200,000/2 years

07/01/2022-
6/30/2024 Principal Investigator
Investigation of the interaction among structural, functional, and proteinopathies networks in MCI and AD patients via a contrastive learning-based, multigraph representation framework on the multi-modal neuroimaging data of MRI, fMRI, and PET modalities. The proposed framework will be used to identify and evaluate a multi-modal image biomarker for AD conversion in the MCI population from a multi-site dataset.

Current

“Deep Learning-Based Phenotyping and Treatment Optimization of Heart Failure with Preserved Ejection Fraction”

03/15/2022-
02/28/2026 NIH R01, 1R01HL159183

Senior/Key Person (PI: Quanzheng Li)

Development of a multi-modal deep learning model on combined imaging and EHR data for the purpose of holistic HFpEF patient portrayal, disease phenotyping, as well as treatment optimization.

“Simulated Computed Tomography Scans Generated from X-Rays Using a Diffusion Model”

10/01/2024-
09/30/2026 American Foundation for Surgery of the Hand (AFSH), Direct cost: \$60,000/2 years

Co-PI (PI: Neal Chen)

Developing an AI model that can generate near-real wrist CTs from X-ray images based on the conditional diffusion model.

Report of Local Teaching and Training

Research Supervisory and Training Responsibilities:

2018-	Supervised visiting students, graduate research interns, and research fellows (average 2-3 per year).	Massachusetts General Hospital. Two hours of lab meetings per week; One hour per week per student and research fellow for 1:1 discussion and supervision.
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Other Mentored Trainees and Faculty:

2019-2023	Hui Ren, Research Fellow at MGH. Mentoring role: Advising research on cardiac image analysis and Electronic Health Record (EHR) modeling. Accomplishment: Research presented at the 2019 <i>Annual Meeting of American Heart Association</i> , 2019 <i>IEEE International Symposium on Biomedical Imaging</i> , 2022 <i>International Workshop on Advances in Simplifying Medical Ultrasound</i> , and 2023 <i>AMIA Clinical Informatics Conference</i> .
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2019-2020	Haixing Dai, Ph.D. student at the University of Georgia Graduate research intern at MGH. Mentoring role: Advising research on graph-based deep learning models for brain network modeling from MEG and PET images.
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2021-2023	Jerome Charton, Research Fellow at MGH. Mentoring role: Advising research on the computational modeling of ultrasound, especially echocardiogram images. Accomplishment: Research presented at the 2022 <i>International Workshop on Advances in Simplifying Medical Ultrasound</i> .
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2022-	Sekeun Kim, Research Fellow at MGH. Mentoring role: Advising research on the differential diagnosis for cardiac conditions and image modeling for echocardiography.
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2023- Siyeop Yoon, Research Fellow at MGH. **Mentoring role:** Advising research on the generative modeling for medical images utilizing diffusion models.

Local Invited Presentations:

Report of Regional, National and International Invited Teaching and Presentations

☒ *No presentations below were sponsored by 3rd parties/outside entities*

Regional

2024 “Utilizing Large Language Models for Effective Management and Analysis of Nuclear Medicine Radiology Reports” (Invited Talk)
Hampton Symposium 2024, Boston, MA

2025 “Engineering Large Pre-trained Models: Techniques and Best Practices” (Invited Talk)
School Of Engineering And Applied Sciences, Harvard University

National

2017 “Towards Practical Problems in Deep Learning for Radiology Image Analysis” (Invited Talk)
Nvidia GPU Technology Conference, San Jose, CA.

2018 “Deep Learning Algorithm for rapid automatic detection of pneumothorax on chest CT” (selected oral abstract)
Annual Meeting of American Roentgen Ray Society, Washington, D.C.

2019 “Personalized Healthcare for Heart Failure with Preserved Ejection Fraction (HFpEF): Diagnosis, Phenotyping and Treatment Optimization with Imaging and EHR Data” (Invited Talk)
Department of Statistics, The University of Georgia, Athens, GA.

2022 “Novel Methodologies for Combined Image and EMR Modeling” (Invited Talk)
Department of Computer Science and Engineering, The University of Texas at Arlington, Arlington, TX

2022 “Data Governance of the SAGES CVS Challenge” (Invited Talk)
Society of American Gastrointestinal and Endoscopic Surgeons, Houston, TX (Virtual)

2024 “Development of Specialized Large Language Models for Radiology Report Processing” (Invited Talk)
31st Annual Council on Ionizing Radiation Measurements & Standards Meeting, Rockville, MD.

2024 “Understanding Neurodegenerative Diseases via Multi-modal Data Analytics, Generative Modeling, and Counterfactual Causal Inference” (Invited Talk)
The New Investigators in Alzheimer’s Disease Meeting, Bethesda, MA

International

2019 “Automated Segmentation of Cervical Nuclei in Pap Smear Images using Deformable Multi-path Ensemble Model” (selected oral full-length paper)
IEEE International Symposium on Biomedical Imaging, Venice, Italy.

2023 “Identification of Causal Relationship between Amyloid-beta Accumulation and Alzheimer’s Disease Progression via Counterfactual Inference”
Optica Imaging Congress, Boston, MA (selected oral abstract)

2023 “Recent Advances in Sparse and Ultra-Sparse Reconstruction and Generative Modeling for Medical Imaging”
Optica Imaging Congress, Boston, MA (invited talk)

2024 “Cross-Modal Retrieval for Alzheimer's Disease Diagnosis Using CLIP-Enhanced Dual Deep Hashing”
and
“Conditional Score-Based Diffusion Model for Cortical Thickness Trajectory Prediction”
AMIA 2024 Annual Symposium, San Francisco, CA (selected oral abstract)

Report of Technological and Other Scientific Innovations

Automatic pre-screening method for pneumothorax detection, 2017	Development of a software system for automatically detecting pneumothorax from CT images. The system has been validated internally by multiple radiologists. The innovation was filed as an Invention Disclosure to Partners HealthCare in 2017.
Inference of 3D chest CT images from 2D chest radiographs, 2019	Development of an invention to directly infer 3D chest CT images from 2D radiographs. The technique learns the mapping between the 2D image and the 3D spatial structure information of the corresponding CT image from the teaching dataset, thus achieving trans-dimensional 2D-3D mapping. The innovation was filed as an Invention Disclosure to Partners HealthCare in 2020.
COVID-19 risk score prediction system and its integration into clinical workflow, 2020	Developing a risk assessment model for COVID-19 based on deep learning-based analysis of EHR and chest radiography data in the Emergency Department and facilitating deployment and integration of the model into the clinical workflow via multiple medical informatics solutions (hospital FHIR infrastructure, Epic Cognitive Computing, Nvidia Clara). The system is currently running online, making >4,000 inferences per day.
Predicting Alzheimer's Disease by Hierarchical Graph Convolution from Positron Emission Tomography Imaging (PETNet), 2021	Developing a Graph Convolutional Network-based Positron Emission Tomography (PET) image analysis program PETNet. The program can automatically diagnose Alzheimer's Disease (AD) and Mild Cognitive Impairment (MCI) from the input PET imaging data. The program features a hierarchical graph inference algorithm for fine-grained analysis of large-scale voxel-wise data and a Neural Architecture Search (NAS) scheme for identifying the optimal graph representing the brain images. The invention has been filed and approved as an Innovation Disclosure to MGB.
Machine Learning Model for the Prediction of Early Discharge of Patient Underwent Transcatheter Aortic Valve Replacement (TAVR) Using Electronic Medical Record, 2023	Developing a machine learning algorithm to predict the early discharge (<36h) of patients who underwent transcatheter aortic valve replacement, given in-house electronic medical record (EMR) information. The prediction is based on data extracted from EMR during a patient's hospital stay, including demographics, vital signs, medical history, related laboratory results, and medications. The invention has been filed for an Innovation Disclosure to MGB.
AUTOMATED DETECTION AND MANAGEMENT OF VALVULAR HEART DISEASE USING MACHINE LEARNING	Developing an Aortic Stenosis Ensemble Risk Prediction (AS-ERP) Model. AS-ERP performs Aortic Stenosis patient outcome (length of stay and readmission) prediction based on the input Electronic Medical Records (EMR) data. The model utilizes an ensemble learning scheme consisting of three machine learning classifiers for patient outcome prediction. Internal validation performance meets the clinical acceptance criteria and is superior to the current risk score system developed by the Society of Thoracic Surgeons. US Patent No. 2024/0379239

Report of Scholarship

Peer-Reviewed Scholarship in print or other media:

Research Investigations *indicates co-first authorship, **indicates my mentee

- 1 A Zhong*, **X Li***, D Wu, H Ren**, K Kim, Y Kim, V Buch, N Neumark, B Bizzo, WY Tak, SY Park, YR Lee, MK Kang, JG Park, BS Kim, WJ Chung, N Guo, I Dayan, MK Kalra, Q Li. Deep metric learning-based image retrieval system for chest radiograph and its clinical applications in COVID-19. *Medical Image Analysis*. 2021;70:101993.
- 2 I Dayan, HR Roth, A Zhong, A Harouni, A Gentili, AZ Abidin, A Liu, AB Costa, BJ Wood, CS Tsai, CH Wang, CN Hsu, CK Lee, P Ruan, D Xu, D Wu, E Huang, FC Kitamura, G Lacey, GC de Antônio Corradi, G Nino, HH Shin, H Obinata, H Ren**, JC Crane, J Tetreault, J Guan, JW Garrett, JD Kaggie, JG Park, K Dreyer, K Juluru, K Kersten, MAB Rockenbach, MG Linguraru, MA Haider, M AbdelMaseeh, N Rieke, PF Damasceno, PMC e Silva, P Wang, S Xu, S Kawano, S Sriswasdi, SY Park, TM Grist, V Buch, W Jantarabenjakul, W Wang, WY Tak, **X Li**, X Lin, YJ Kwon, A Quraini, A Feng, AN Priest, B Turkbey, B Glicksberg, B Bizzo, BS Kim, C Tor-Díez, CC Lee, CJ Hsu, C Lin, CL Lai, CP Hess, C Compas, D Bhatia, EK Oermann, E Leibovitz, H Sasaki, H Mori, I Yang, JH Sohn, KN Murthy, LC Fu, MRF de Mendonça, M Fralick, MK Kang, M Adil, N Gangai, P Vateekul, P Elhajjar, S Hickman, S Majumdar, SL McLeod, S Reed, S Gräf, S Harmon, T Kodama, T Puthanakit, T Mazzulli, VL de Lavor, Y Rakvongthai, YR Lee, Y Wen, FJ Gilbert, MG Flores, Q Li. Federated learning for predicting clinical outcomes in patients with COVID-19. *Nature Medicine*. 2021;27:1735.
- 3 L Manubens-Gil, Z Zhou, H Chen, A Ramanathan, X Liu, Y Liu, A Bria, T Gillette, Z Ruan, J Yang, M Radojević, T Zhao, L Cheng, L Qu, S Liu, K E Bouchard, L Gu, W Cai, S Ji, B Roysam, C-W Wang, H Yu, A Sironi, D M Iascone, J Zhou, E Bas, E Conde-Sousa, P Aguiar, **X Li**, Y Li, S Nanda, Y Wang, L Muresan, P Fua, B Ye, H-y He, J F Staiger, M Peter, D N Cox, M Simonneau, M Oberlaender, G Jefferis, K Ito, P Gonzalez-Bellido, J Kim, E Rubel, H T Cline, H Zeng, A Nern, A-S Chiang, J Yao, J Roskams, R Livesey, J Stevens, T Liu, C Dang, Y Guo, N Zhong, G Tourassi, S Hill, M Hawrylycz, C Koch, E Meijering, G A Ascoli, H Peng. BigNeuron: a resource to benchmark and predict performance of algorithms for automated tracing of neurons in light microscopy datasets. *Nature Methods*. 2023;20:824.
- 4 J Holmes, Z Liu**, L Zhang, Y Ding, T Sio, L McGee, J Ashman, **X Li**, T Liu, J Shen, W Liu. Evaluating large language models on a highly-specialized topic, radiation oncology physics. *Frontiers in Oncology*. 2023;13.
- 5 O Shen, J Pratap, **X Li**, N Chen, A Bhashyam. How Does ChatGPT Use Source Information Compared with Google? A Text Network Analysis of Online Health Information. *Clinical Orthopaedics and Related Research*. 2024; 482(4):578.
- 6 S Kim**, H Ren**, J Charton, J Hu, C Gonzalez, J Khambhati, J Cheng, J DeFrancesco, A Waheed, S Marciniak, F Moura, R Cardoso, B Lima, S McKinney, M Picard, **X Li**, Q Li. Assessment of Valve Regurgitation Severity via Contrastive Learning and Multi-view Video Integration. *Physics in Medicine & Biology*. 2024; 69(4):045020.
- 7 Z Chen, H Ren, Q Li, **X Li**. Motion Correction and Super-Resolution for Multi-slice Cardiac Magnetic Resonance Imaging via an End-to-End Deep Learning Approach. *Computerized Medical Imaging and Graphics*. 2024; 115:102389.
- 8 Z Liu, L Zhang, Z Wu, X Yu, C Cao, H Dai, N Liu, J Liu, W Liu, Q Li, D Shen, **X Li**, D Zhu, T Liu. Surviving ChatGPT in Healthcare. *Frontiers in Radiology*. 2024; 3:1224682.
- 9 H Dai, Z Liu, W Liao, X Huang, Y Cao, Z Wu, L Zhao, S Xu, F Zeng, W Liu, N Liu, S Li, D Zhu, H Cai, L Sun, Q Li, D Shen, T Liu, **X Li**. AugGPT: Leveraging ChatGPT for Text Data Augmentation. *IEEE Transactions on Big Data*. 2024; In Press.

- 10 Z Wu, L Zhang, C Cao, Z Liu, L Zhao, Y Li, X Yu, H Dai, C Ma, G Li, W Liu, Q Li, D Shen, **X Li**, D Zhu, T Liu. Exploring the Trade-Offs: Unified Large Language Models vs Local Fine-Tuned Models for Highly-Specific Radiology NLI Task. *IEEE Transactions on Big Data*. 2024; In Press.
- 11 K Zhang, R Zhou, E Adhikarla, Z Yan, Y Liu, J Yu, Z Liu, X Chen, B Davison, H Ren, J Huang, C Chen, Y Zhou, S Fu, W Liu, T Liu, **X Li***, Y Chen, L He*, J Zou, Q Li, H Liu, L Sun*. A Generalist Vision-Language Foundation Model for Diverse Biomedical Tasks. *Nature Medicine*. 2024; 30:3129–3141.
- 12 C Chen, J Miao, D Wu, A Zhong, Z Yan, S Kim, J Hu, Z Liu, L Sun, **X Li**, T Liu, P Heng, Q Li. MA-SAM: Modality-agnostic SAM Adaptation for 3d Medical Image Segmentation. *Medical Image Analysis*. 2024; 98:103310.
- 13 T Bradshaw, X Tie, J Warner, J Hu, Q Li, **X Li**. Large language models and large multimodal models in medical imaging: a primer for physicians. *Journal of Nuclear Medicine*. 2025; In Press.
- 14 H Dai**, Y Li**, Z Liu**, L Zhao, Z Wu**, S Song, S Ye, D Zhu, **X Li**, S Li, X Yao, L Shi, T Peng, Q Li, Z Chen, D Zhang, G Mai, T Liu. AD-AutoGPT: An Autonomous GPT for Alzheimer’s Disease Infodemiology. *PLOS Global Public Health*. 2025; In Press.
- 15 S Kim**, P Jin, C Chen, K Kim, Z Lyu, Hui Ren, S Kim, Z Liu**, A Zhong, T Liu, **X Li**, Q Li. MediViSTA: Medical Video Segmentation via Temporal Fusion SAM Adaptation for Echocardiography. *IEEE Journal of Biomedical and Health Informatics*. 2025; In Press.
- 16 Z Chen**, S Kim**, H Ren, S Kim, S Yoon, Q Li, **X Li**. Cine Cardiac Magnetic Resonance Segmentation using Temporal-spatial Adaptation of Prompt-enabled Segment-Anything-Model: A Feasibility Study. *Journal of Cardiovascular Magnetic Resonance*. 2025; In Press.
- 17 Z Liu, Y Li**, P Shu**, A Zhong, H Jinag, Y Pan, L Yang, C Ju, Z Wu, C Ma, C Chen, S Kim**, H Dai**, L Zhao, L Sun, D Zhu, J Liu, W Liu, D Shen, Q Li, T Liu, **X Li**. Radiology-GPT: A Large Language Model for Radiology. *Meta-Radiology*. 2025; In Press.
- 18 S Kim**, P Jin, S Song, C Chen, Y Li, H Ren, **X Li**, Ti Liu, Q Li. EchoFM: Foundation Model for Generalizable Echocardiogram Analysis. *IEEE Transactions on Medical Imaging*. 2025; In Press.
- 19 P Wang, Z Liu, Y Li, J Holmes, P Shu, L Zhang, B Laughlin, DS Toesca, **X Li**, Q Li, S Vora, S Patel, T Sio, T Liu, W Liu. Fine-Tuning Open-Source Large Language Models to Improve Their Performance on Radiation Oncology Tasks. *Medical Physics*. 2025; In Press.

Other peer-reviewed scholarship: *indicates co-first or co-corresponding authorship, **indicates my mentee

Full-length proceedings of meetings which have undergone a formal peer-review process

- 1 J Charton**, H Ren**, J Khambhati, J DeFrancesco, J Cheng, A Waheed, S Marciniak, F Moura, R Cardoso, B Lima, E Steen, E Samset, M Picard, **X Li**, Q Li. View Classification of Color Doppler Echocardiography via Automatic Alignment Between Doppler and B-Mode Imaging. *International Workshop on Advances in Simplifying Medical Ultrasound (ASMUS)* 2022.
- 2 S Rezayi, H Dai, Z Liu**, Z Wu, A Hebbbar, A Burns, L Zhao, D Zhu, Q Li, W Liu, S Li, T Liu, **X Li**. ClinicalRadioBERT: Knowledge-Infused Few-Shot Learning for Clinical Notes Named Entity Recognition. *International Workshop on Machine Learning in Medical Imaging (MLMI)* 2022.
- 3 H Cai, X Huang, Z Liu**, W Liao, H Dai, Z Wu, D Zhu, H Ren**, Q Li, T Liu, **X Li**. Multimodal Approaches for Alzheimer's Detection Using Patients' Speech and Transcript. *Brain Informatics (BI) Conference* 2023.
- 4 Z Liu**, A Zhong, Y Li, L Yang, C Ju, Z Wu, C Ma, P Shu, C Chen, S Kim**, H Dai, L Zhao, D Zhu, J Liu, W Liu, D Shen, Q Li, T Liu, **X Li**. Tailoring Large Language Models to Radiology: A preliminary approach to LLM adaptation for a highly specialized domain. *International Workshop on Machine Learning in Medical Imaging (MLMI)* 2023.
- 5 J Charton**, H Ren**, S Kim**, CM Gonzalez, J Khambhati, J Cheng, J DeFrancesco, A Waheed, S Marciniak, F Moura, R Cardoso, B Lima, M Picard, **X Li**, Q Li. Multi-task Learning for

- Hierarchically-Structured Images: Study on Echocardiogram View Classification. *International Workshop on Advances in Simplifying Medical Ultrasound (ASMUS)* 2023.
- 6 H Dai**, M Hu, Q Li, L Zhang, L Zhao, D Zhu, I Diez, J Sepulcre, F Zhang, X Gao, M Liu, Q Li, S Li, T Liu, **X Li**. Graph-Based Counterfactual Causal Inference Modeling for Neuroimaging Analysis. *International Workshop on Multiscale Multimodal Medical Imaging (MMMI)* 2023.
 - 7 B Kim, J Choi, E Yun, K Kim, **X Li**, Juho Lee. Large-scale Graph Representation Learning of Dynamic Brain Connectome with Transformers. *NeurIPS Workshop on Temporal Graph Learning* 2023.
 - 8 Y Shi, H Ma, W Zhong, Q Tan, G Mai, **X Li**, T Liu, J Huang. Chatgraph: Interpretable Text Classification by Converting ChatGPT Knowledge to Graphs. *IEEE International Conference on Data Mining Workshops (ICDMW)* 2023.
 - 9 S Yoon**, J Pratap, W C Liu, M Tivnan, H Ren, A Bhashyam, Q Li, N Chen, **X Li**. High-resolution 3D CT synthesis from bidirectional X-ray Images using 3D Diffusion Model. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2024.
 - 10 B Kim, J Choi, E Yun, K Kim, **X Li**, J Lee. Learning Dynamic Brain Connectome with Graph Transformers for Psychiatric Diagnosis Classification. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2024.
 - 11 Z Chen**, H Ren, Q Li, **X Li**. Motion Correction and Super-Resolution for Multi-slice Cardiac Magnetic Resonance Imaging via a Multi-stage Deep Learning Approach. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2024.
 - 12 J Pratap, S Yoon**, W C Liu, Q Li, A Bhashyam, N Chen, **X Li**. Zero-Shot Novel View Synthesis of Wrist X-Rays using Latent Diffusion Model. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2024.
 - 13 Y Huang, L Sun, H Wang, S Wu, Q Zhang, Y Li, C Gao, Y Huang, W Lyu, Y Zhang, X Li, H Sun, Z Liu, Y Liu, Y Wang, Z Zhang, B Vidgen, B Kailkhura, C Xiong, C Xiao, C Li, E P Xing, F Huang, H Liu, H Ji, H Wang, H Zhang, H Yao, M Kellis, M Zitnik, M Jiang, M Bansal, J Zou, J Pei, J Liu, J Gao, J Han, J Zhao, J Tang, J Wang, J Vanschoren, J Mitchell, K Shu, K Xu, K-W Chang, L He, L Huang, M Backes, N Z Gong, P S Yu, P-Y Chen, Q Gu, R Xu, R Ying, S Ji, S Jana, T Chen, T Liu, T Zhou, W Y Wang, **X Li**, X Zhang, X Wang, X Xie, X Chen, X Wang, Y Liu, Y Ye, Y Cao, Y Chen, Y Zhao. Position Paper: TrustLLM: Trustworthiness in Large Language Models. *International Conference on Machine Learning (ICML)* 2024.
 - 14 S Yoon**, R Hu, Y Wang, M Tivnan, Y D Son, D Wu, **X Li**, K Kim, Q Li. Efficient Volumetric Conditional Score-based Residual Diffusion Model for PET Denoising. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)* 2024.
 - 15 M Tivnan, S Yoon**, Z Chen, **X Li**, D Wu, Q Li. Hallucination Index: An Image Quality Metric for Generative Reconstruction Models. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)* 2024.
 - 16 Y Shi**, S Xu, T Yang, Z Liu, T Liu, **X Li***, N Liu*. MKRAG: Medical Knowledge Retrieval Augmented Generation for Medical Question Answering. *Annual Symposium of American Medical Informatics Association (AMIA)* 2024.
 - 17 Z Chen, S Kim, H Ren, Q Li, **X Li**. Temporal-spatial Adaptation of Promptable SAM Enhances Accuracy and Generalizability of Cine CMR Segmentation. *International Workshop on Foundation Models for General Medical AI (MedAGI)* 2024.
 - 18 H Cui, L Mao, X Liang, J Zhang, H Ren, Q Li, **X Li***, C Yang*. Biomedical Visual Instruction Tuning with Clinician Preference Alignment. *Annual Conference on Neural Information Processing Systems (NeurIPS)* 2024.
 - 19 Y Pan, H Jiang, J Chen, Y Li, H Zhao, Y, P, Z Wu, Z Liu**, D Zhu, **X Li**, Y Abate, T Liu. EG-SpikeFormer: Eye-Gaze Guided Transformer on Spiking Neural Networks for Medical Image Analysis. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2025.

- 20 Y Shi**, Q Li, J Sun, **X Li***, N Liu*. Enhancing Cognition and Explainability of Multimodal Foundation Models with Self-Synthesized Data. *International Conference on Learning Representations (ICLR)* 2025.
- 21 Y Li**, S Kim, Z Wu, H Jiang, Y Pan, P Jin, S Song, Y Shi, X Yu, T Yang, T Liu, Q Li, **X Li**. ECHOPulse: ECG Controlled Echocardiogram Video Generation. *International Conference on Learning Representations (ICLR)* 2025.
- 22 Y Oh, P Jin, S Park, S Kim**, S Yoon**, J Kim, K Kim, **X Li***, Q Li*. Distribution-aware Fairness Learning in Medical Image Segmentation from a Control-Theoretic Perspective. *International Conference on Machine Learning (ICML)* 2025.
- 23 S Yoon, Y Oh, **X Li**, Y Xin, M Cereda, Q Li. High-fidelity 3D Lung CT Synthesis in ARDS Swine Models using Score-based 3D Residual Diffusion Models. *SPIE Medical Imaging* 2025.
- 24 S Yoon, S Song, P Jin, M Tivnan, Y Oh, S Kim, D Wu, **X Li**, Q Li. Cascaded 3D Diffusion Models for Whole-body 3D 18-FDG PET/CT synthesis from Demographics. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)* 2025.
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Title: **Machine Learning Approaches towards Holistic Brain Functional Space Discovery from fMRI Big Data**

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Advisor: Distinguished Prof. Tianming Liu

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Abstracts, Poster Presentations and Exhibits Presented at Professional Meetings:

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